



Turning Analytics into Timely Action – Leveraging DIEM household data

Leveraging Evidence for Effective
Anticipatory Action

20 April 2026

Office of Emergencies and Resilience
Food and Agriculture Organization of the United Nations (FAO)

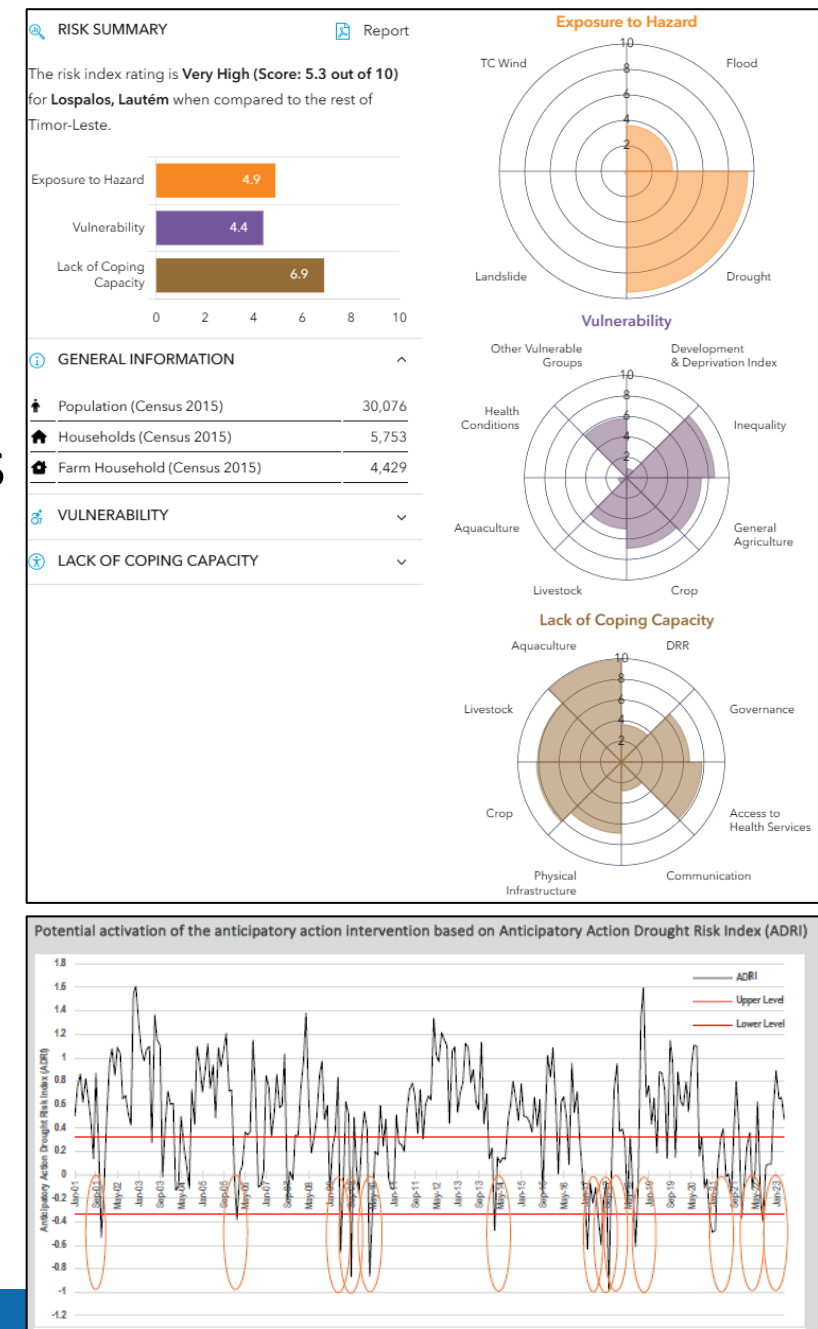
The concept

- Risk of agriculture and food security degradation due to shock(s)
- Shocks do not come in isolation:
 - Some make others more likely
 - Some shock interactions exacerbate the effect of other shocks
 - Some may reduce the effect of other shocks



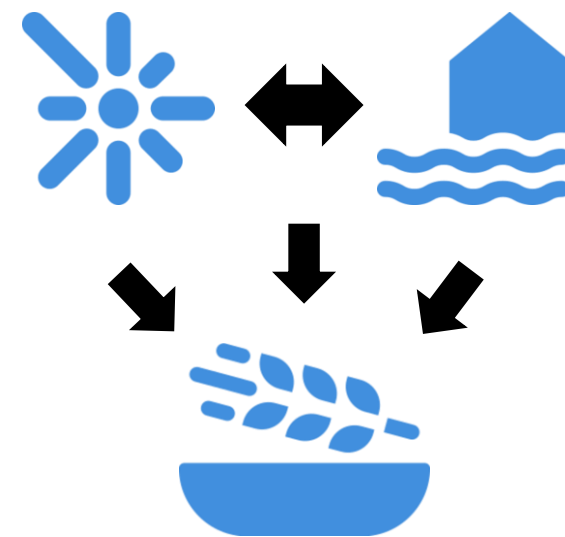
The gap

- AA indicators weights, thresholds, and triggers mainly based on assumptions and expert judgement
- Lack of empirical analysis on **how** historical shocks affected agricultural livelihoods and food security (shock magnitude, severity of impacts, geographic scope, seasonality, lag, shocks combinations, cascading effects, buffers etc.)



The logic

- Strengthening expert judgement by assessing empirically the cumulative and interdependent impacts of multiple historical shocks on household food security to predict how upcoming shock signals could result in potential food insecurity
- Harnessing DIEM historical household dataset enriched with global geocoded datasets on shocks



The ingredients - DIEM

Coverage

5+ years of data
30+ countries
800,000+ household interviews

1

Sample

Representative at subnational level
Geolocalized at adm2

2

Questionnaire

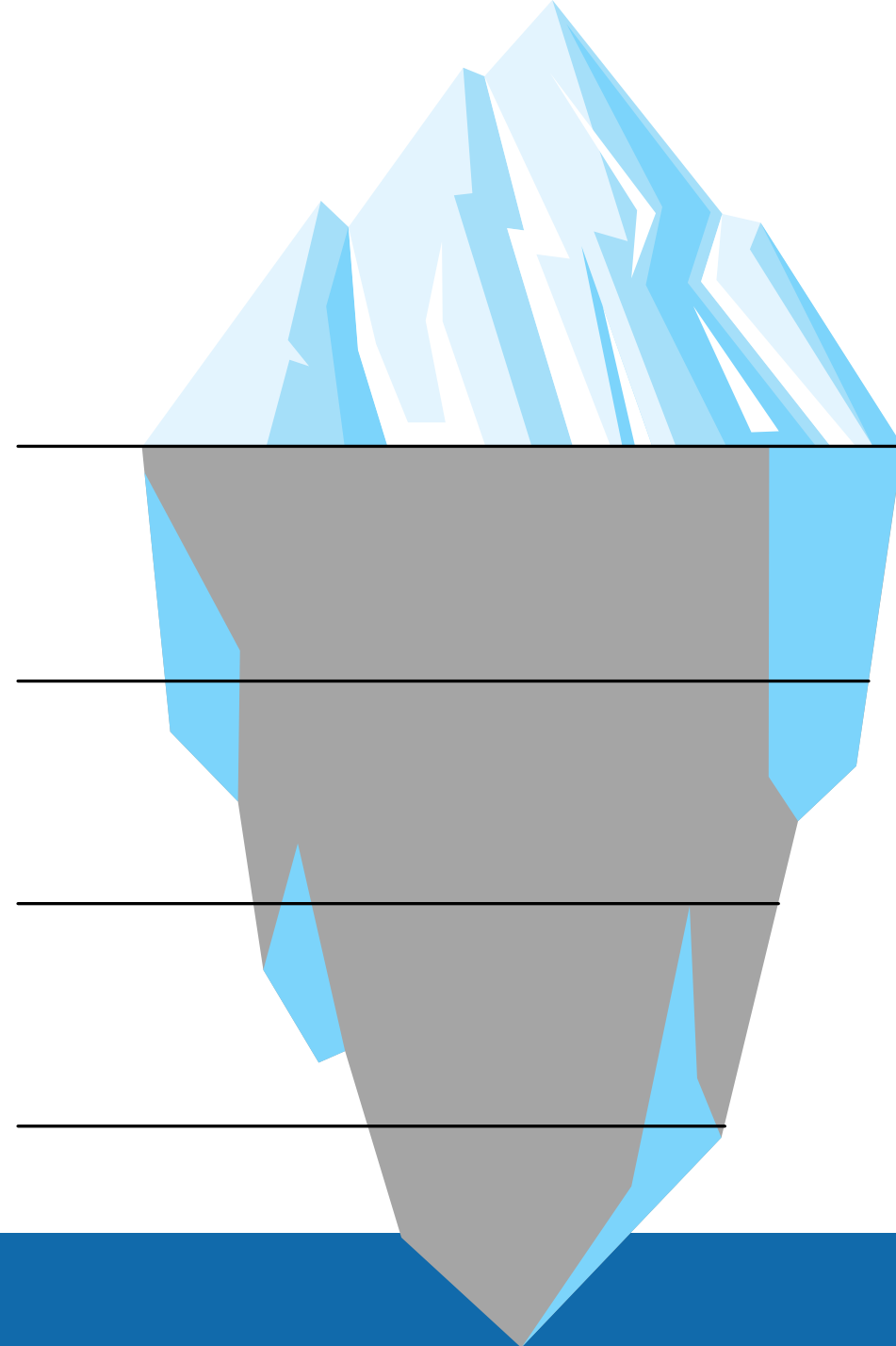
Comparable across countries / time

3

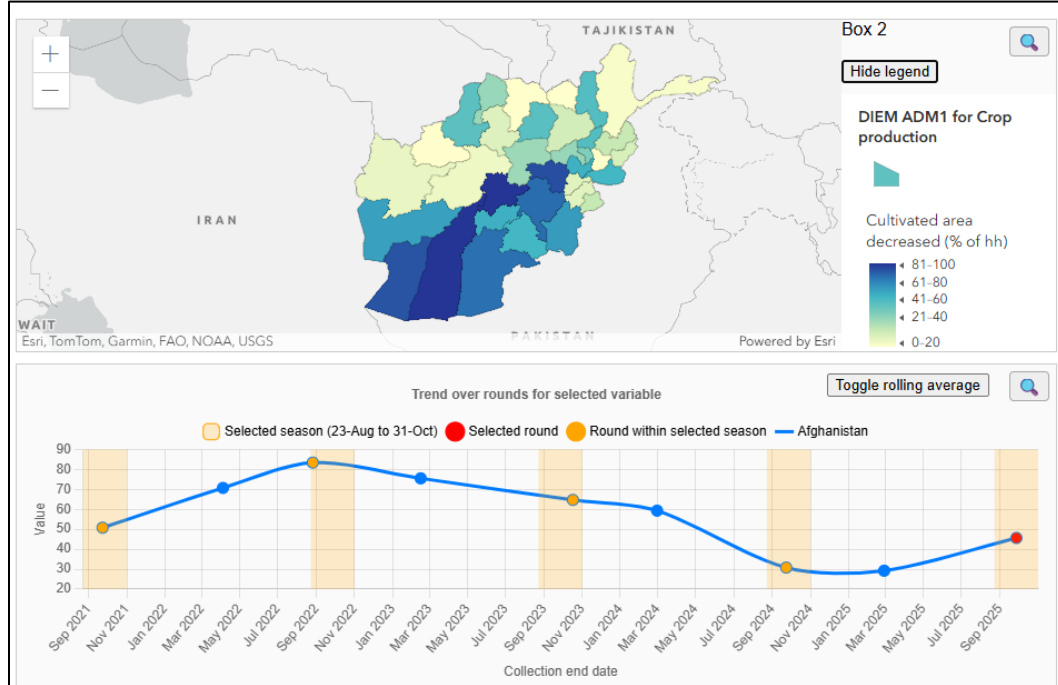
Hub

Accessibility, transparency,
continuity

4



The ingredients - DIEM



Shocks: self-reported, endogeneity, need for more objective measurement

SOCIO-DEMOGRAPHIC FEATURES

- Gender, age
- Displacement
- Wealth proxies

SHOCKS & INCOMES

- Shocks*
- Income ranking
- Income change

CROPS

- Main product
- Change in production
- Production difficulties
- Changes in sales & prices
- Marketing difficulties

LIVESTOCK

FISHERIES/AQUACULTURE

FOOD SECURITY OUTCOMES

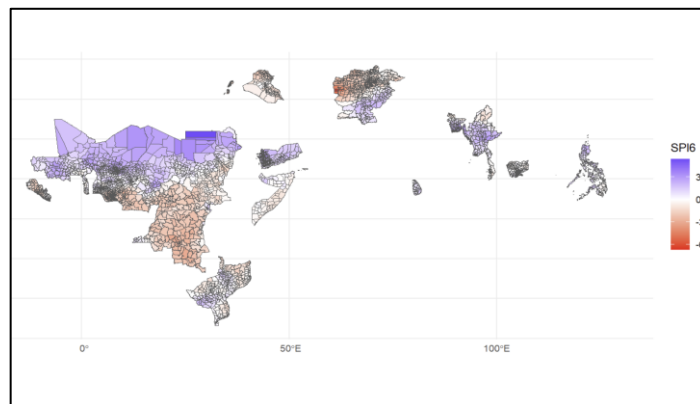
- Consumption (e.g. FCS)
- Experience (e.g. FIES)
- Coping strategies (e.g. LCSi)

NEEDS FOR ASSISTANCE

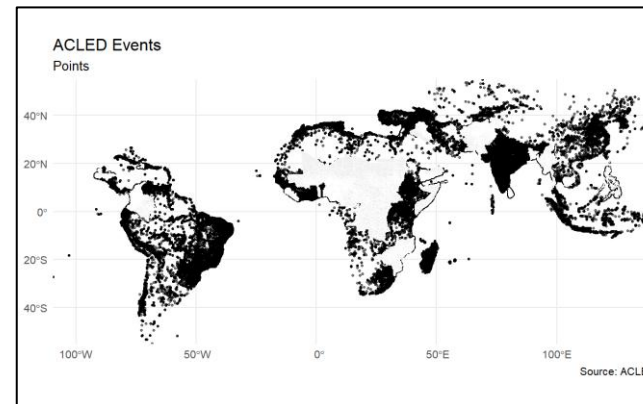
- Household needs
- Assistance received

The ingredients – Global datasets on shocks

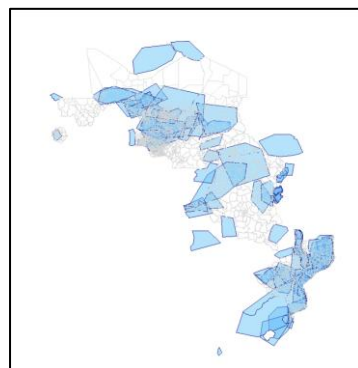
Drought (SPI)



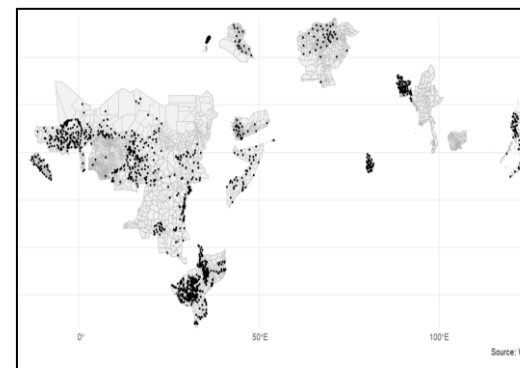
Conflict (ACLED)



Flood (DFO, GLOFAS, EVE)



Prices (WFP, GIEWS)



The recipe

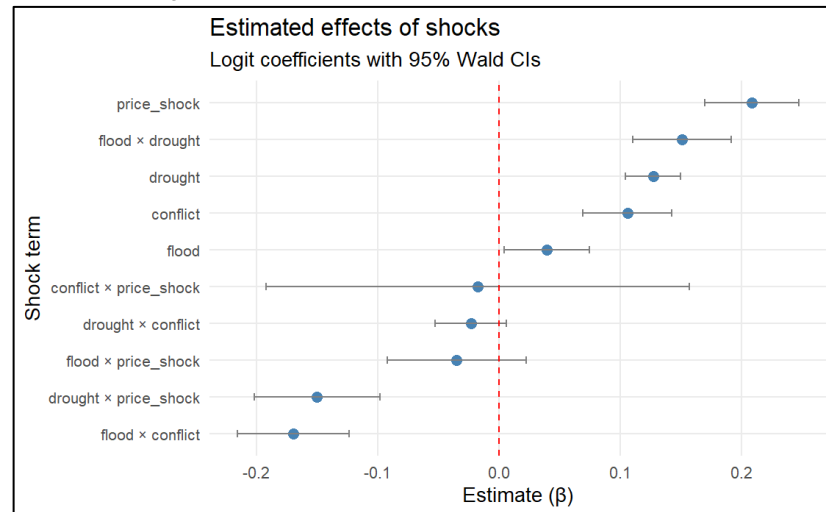
The **Multi-Shock Index (MSI)** estimates:

- The likely change in food security due to shocks and shock interactions at the adm2 level
- The risk of correlated shocks likely to stem from the primary shock

Logistic regression (outcome: FIES)
1) cross-country coefficients, 2)
country with adm1 fixed effects

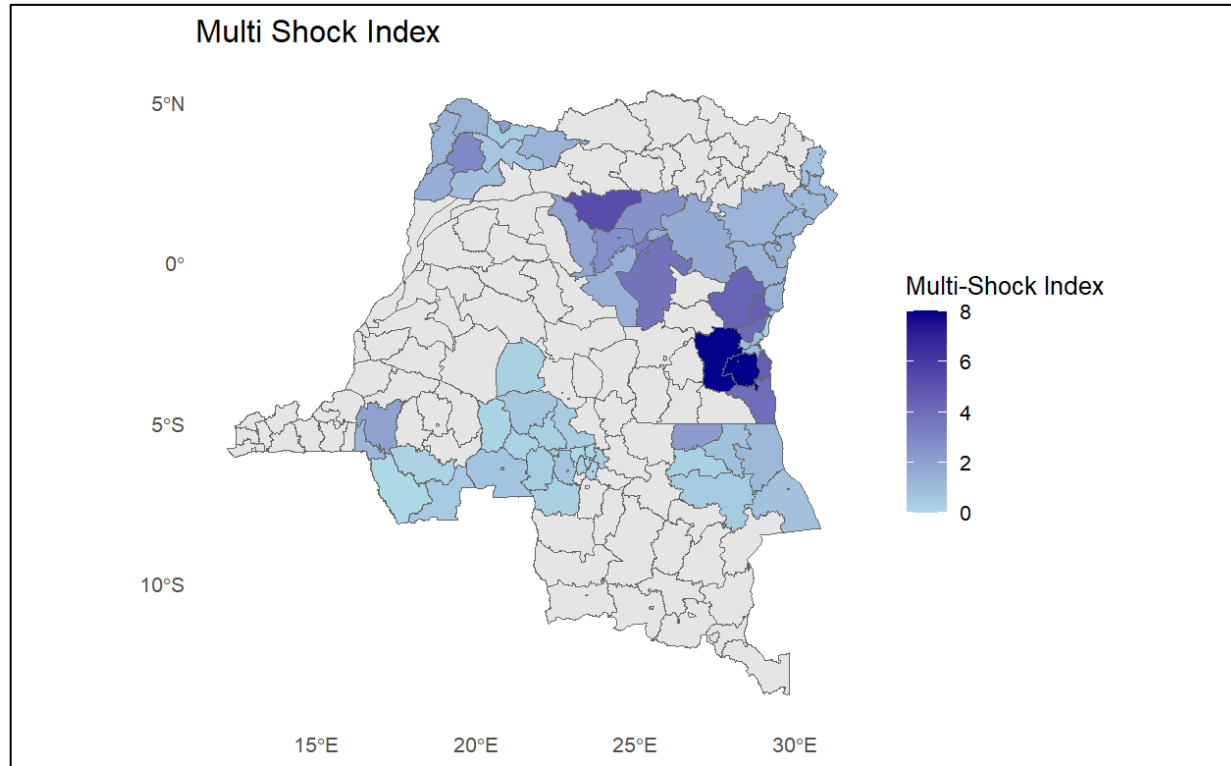


MSI = marginal probabilities from coefficients
x sum of 10 year probability of shock and
shock interaction occurrence at adm2

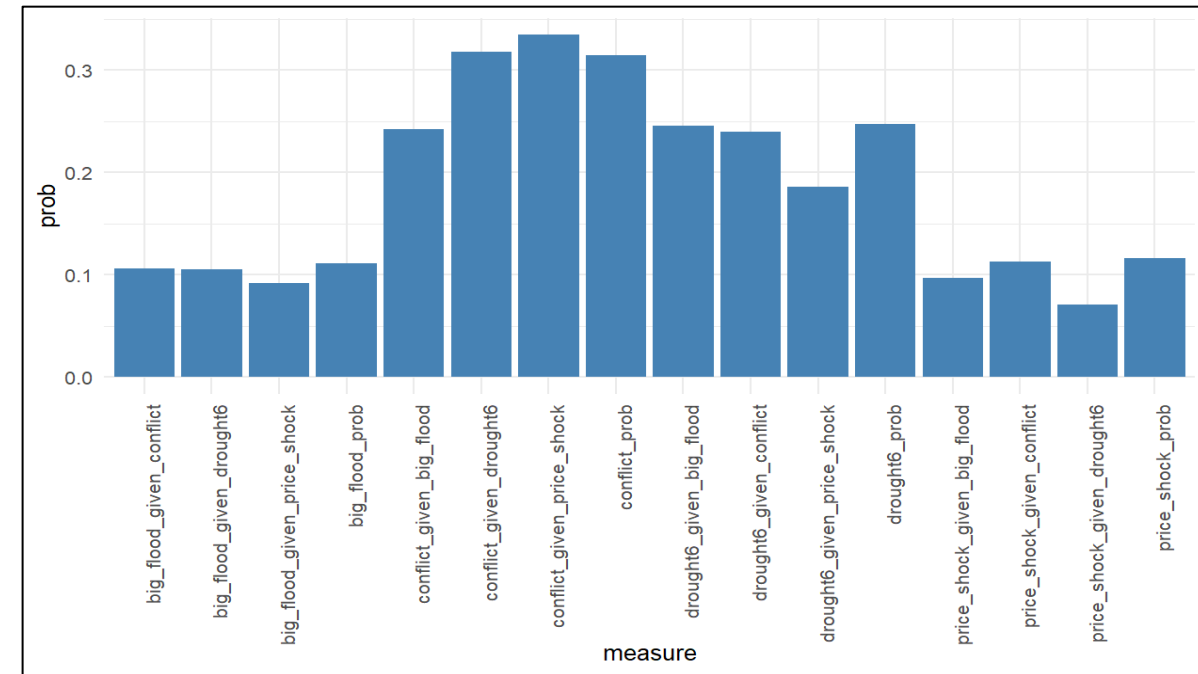


adm2_name	FI_baseline	P_drought	P_drought_conflict	delta_drought_conflict	P_flood_conflict	delta_flood_conflict	delta_season	MSI_index
Kahemba	60.57	63%	0%	1.11	0%	8.56	9.97	0.50
Feshi	60.57	52%	4%	1.11	0%	8.56	9.97	0.17
Popokabaka	60.57	26%	0%	1.11	7%	8.56	9.97	1.22
Kenge	60.57	33%	7%	1.11	14%	8.56	9.97	2.04
Budjala	59.35	49%	5%	1.12	1%	8.68	10.13	0.90
Gemena	59.35	67%	6%	1.12	20%	8.68	10.13	2.90
Zongo	59.35	58%	3%	1.12	3%	8.68	10.13	0.93
Libenge	59.35	63%	5%	1.12	5%	8.68	10.13	1.28
Kungu	59.35	40%	7%	1.12	8%	8.68	10.13	1.56
Yakoma	67.49	60%	5%	1.01	8%	7.68	8.92	1.38
Bosobolo	67.49	88%	0%	1.01	0%	7.68	8.92	1.36
Gbadolite	67.49	57%	5%	1.01	10%	7.68	8.92	2.27
Mobayi-Mbongo	67.49	45%	0%	1.01	0%	7.68	8.92	0.52
Businga	67.49	52%	0%	1.01	0%	7.68	8.92	0.65

The output

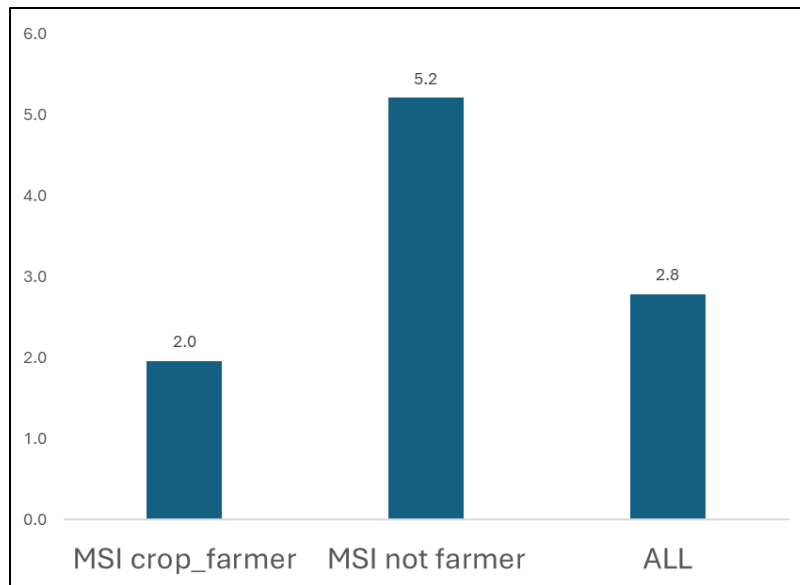
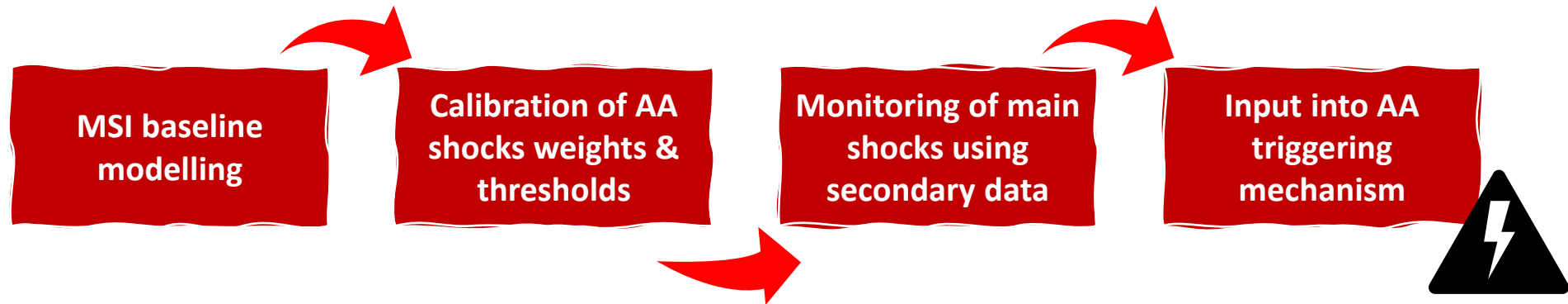


DRC multi-shock index



Risk of correlated shock j
associated with shock i

How could it be applied to AA



AA targeting
(MSI by population group)

The background of the slide is a solid blue color. Overlaid on this background is a world map. The map is created using a halftone or dot-matrix pattern, where the density of the dots varies to represent the shapes of the continents. The dots are a slightly lighter shade of blue than the background.

Thanks!

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